Grade 8
Unit 3
Lesson 8 Practice

Name___Answer Key_______
Date______________________
Period_____________________

1. Circle the significant digits in each number. Place a check in the column(s) to reflect which rule(s) you used when making your decision.

	Non-Zero Digits Always Significant	Zeroes Between Non-Zero Digits Always Significant	Zeroes to the Left of Non-Zero Digits Not Significant	All Zeroes to the Right are Significant if There is a Decimal Point
0.8709	✓	✓	✓	
17,062	✓	✓		
1,480	✓			
6.2	✓			
819.0	✓			✓

Use the given information to answer questions 2 and 3.

The student loan debt in the year 2020, was \$1.58 trillion. The population in California in 2020 was estimated to be 39.5 million.

2. If the people living in California were to pay off student loan debt, how much would each person pay?

$\frac{1{,}580{,}000{,}000{,}000}{39{,}500{,}000} = \$40{,}000$

3. The population of San Francisco in 2020 was 8.73 million. If the people living in San Francisco were to pay off the student loan debt, how much would each person pay?

$\frac{1{,}580{,}000{,}000{,}000}{8{,}730{,}000} = \$180{,}985.11$

4. Universal Studios' Volcano Bay uses 3.6 million gallons of water. Disney's Typhoon Lagoon uses 3.0 million gallons of water. How much water is used between the two parks? Write your answer in scientific notation.

6.6×10^6

$3{,}600{,}000 + 3{,}000{,}000 = 6{,}600{,}000$

Use the given information to answer questions 5-7.

The approximate distance between Neptune and the sun is 4.5×10^9. The approximate distance between Mercury and the sun is 5.79×10^7.

5. What is the difference in distances between the two planets and the sun? Write your answer in scientific notation.

$4.5 \times 10^9 - 5.79 \times 10^7$
$450 \times 10^7 - 5.79 \times 10^7 = 444.21 \times 10^7$
4.4421×10^9

6. What is the sum in distances between the two planets and the sun? Write your answer in scientific notation.

$4.5 \times 10^9 + 5.79 \times 10^7$
$4.5 \times 10^9 + 0.0579 \times 10^9$
4.5579×10^9

7. How many times greater is the distance between Neptune and the sun than Mercury and the sun?

$\frac{4.5 \times 10^9}{5.79 \times 10^7} = \frac{4.5}{5.79} \times 10^2$ Approximately 0.77×10^2 or 7.7×10 or 77 times

Use the given information to answer questions 8 and 9.

The approximate area of Rhode Island, in square miles, is 4×10^3. The approximate area of Washington, D.C., in square miles, is 1.77×10^2.

8. How much smaller is Washington, D.C. is than Rhode Island? Write your answer in scientific notation?
$4 \times 10^3 - 1.77 \times 10^2$
$4 \times 10^3 - .177 \times 10^3$ $= 3.823 \times 10^3$ square miles smaller

9. What is the total area of Rhode Island and Washington, D.C.?
$4 \times 10^3 + 1.77 \times 10^2$
$4 \times 10^3 + .177 \times 10^3$ $= 4.177 \times 10^3$ square miles

10. The recent Georgia Junior National Livestock show measured the average mass of 105 cows shown was 4.305×10^4. At the recent South Carolina Junior National Livestock show, the average mass of 105 cows was 6.25×10^5. What is the difference between the average mass of cows at both shows?
$6.25 \times 10^5 - 4.305 \times 10^4 = 6.25 \times 10^5 - 0.4305 \times 10^5 = 5.8195 \times 10^5$